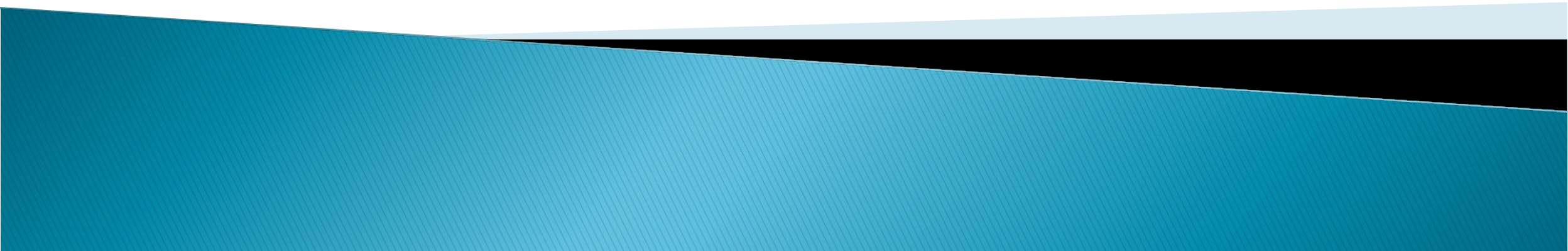
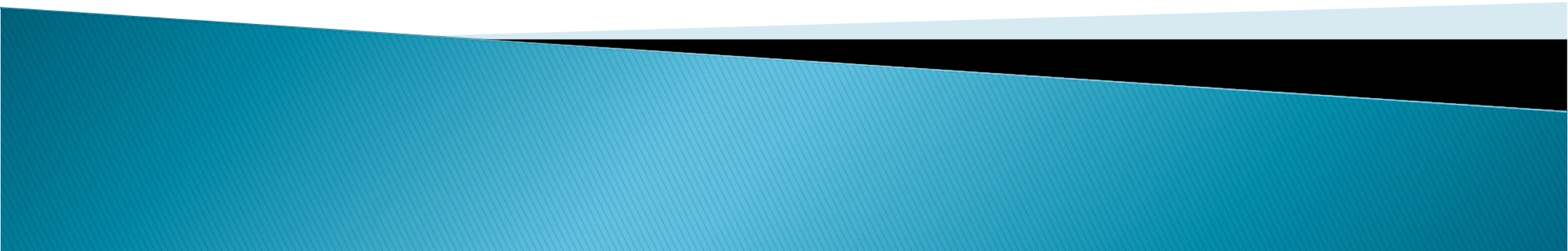


Microsoft Excel



Basic Excel



Filtering



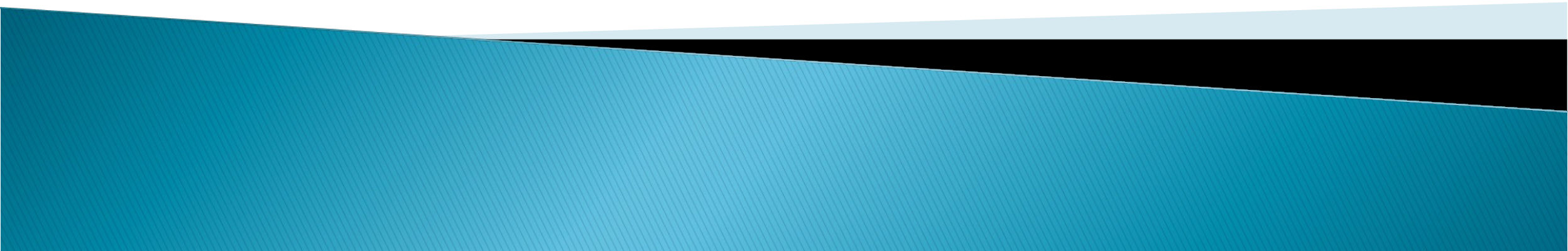
Why Filtering

- We can focus on specific record/rows in a large table of data by setting conditions, so that only specific data is displayed and rest of the records are get hidden temporarily in the spreadsheet.

Filters in MS Excel

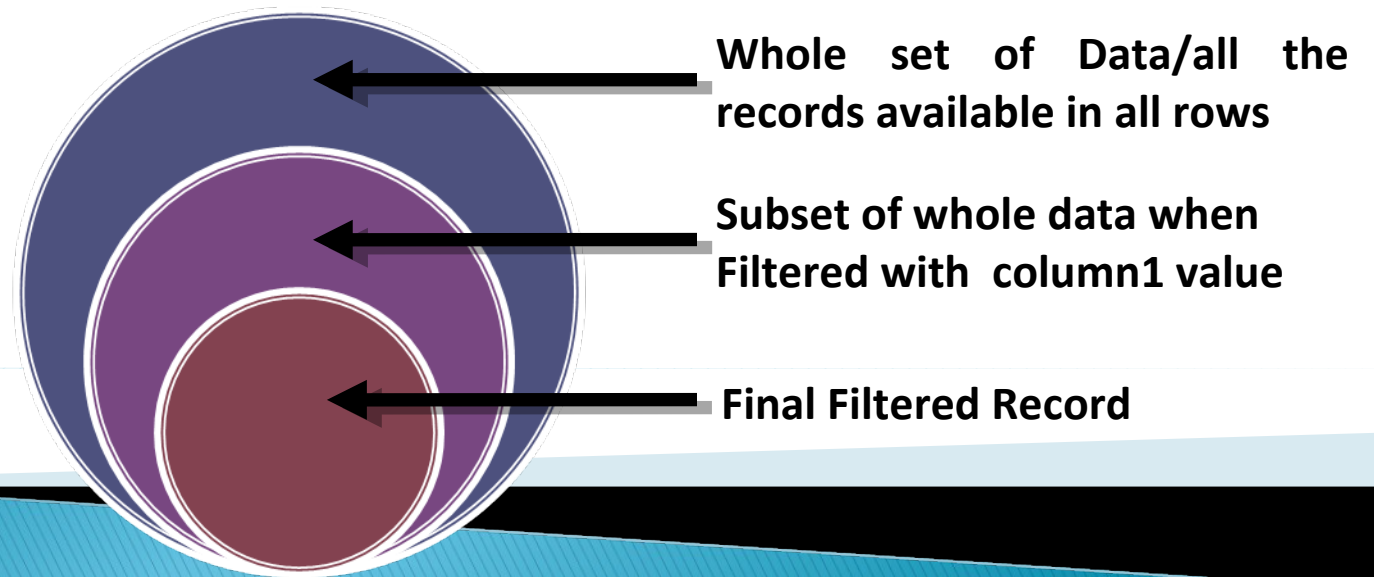
Filtering data in MS Excel refers to displaying only the rows that meet certain conditions. (The other rows get hidden temporarily).

Filter option: **Data → Filter**



Multiple column Filtering

- First we give a condition to a column of a table to get filtered data of whole record.
- Next we give another condition to another column of the same table and so on for getting filtered data(subset of whole record) step by step.



Advanced Filtering

Actions for Advance Filtering:

- Filter the list, In the same place by Criteria
 - Filter the list, Copy to another location Criteria
- 

Text Filter

- ▶ Begins with
 - ▶ Ends with
 - ▶ Contains
 - ▶ Equals
 - ▶ Does not Equal
 - ▶ Does not begin with
 - ▶ Does not end with
- 

Number Filter

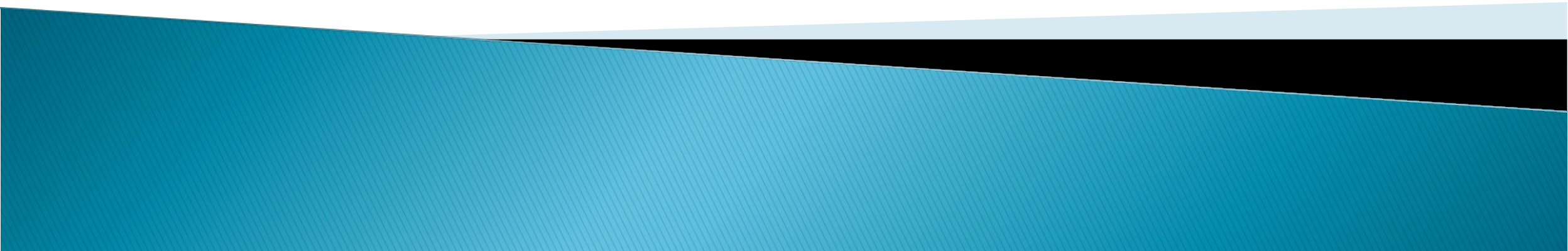
- ▶ Is greater than
 - ▶ Is less than
 - ▶ Top ten
 - ▶ Equals/Does not Equal
 - ▶ Between
 - ▶ Above average
 - ▶ Below average
- 

Data Sorting in Excel



Sorting in MS Excel

Sorting data in MS Excel re-arranges the rows based on the contents of a particular column.

- For numerical data, Sorting could be done in ascending order or descending order.
 - For text data, sorting could be done in alphabetical ascending (A-Z) order or descending order (Z-A).
- 

Three column Sorting

col 1	col 2	col 3
A	10	H
A	10	L
A	20	H
A	20	L
B	15	H
B	15	L
B	12	H
B	12	L

Sort by Col1
Descending

col 1	col 2	col 3
B	15	H
B	15	L
B	12	H
B	12	L
A	10	H
A	10	L
A	20	H
A	20	L

Sort by
Col1,Col2,
Col3
Descending

col 1	col 2	col 3
B	15	H
B	15	L
B	12	H
B	12	L
A	20	H
A	20	L
A	10	H
A	10	L

Sort by Col1
Ascending and
Col2,col3
Descending

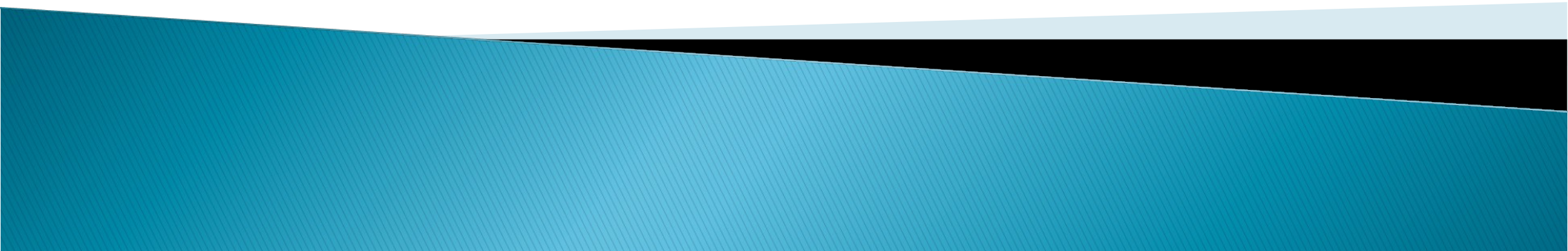
col 1	col 2	col 3
A	20	L
A	20	H
A	10	L
A	10	H
B	15	L
B	15	H
B	12	L
B	12	H

Data Validation in Excel



Data Validation

MS Excel data validation feature allows you to set up certain validation rules that what can be entered into a cell.



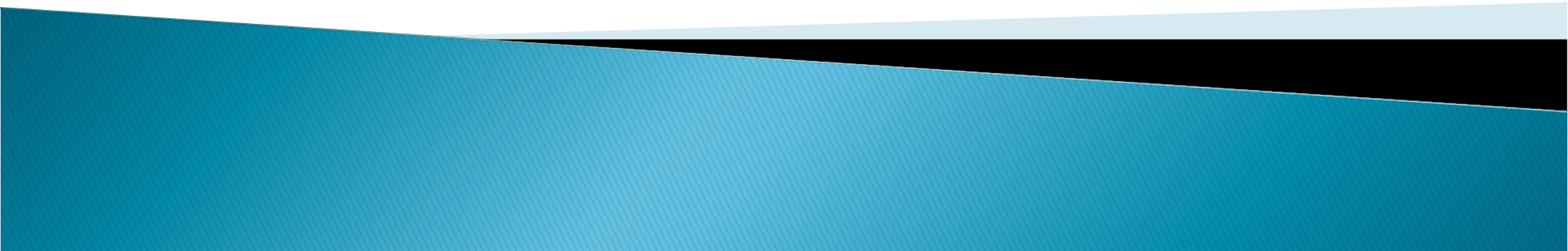
Validation Criteria

To specify the type of data allowable in a cell or range, follow the steps below:

Select the cell or range.

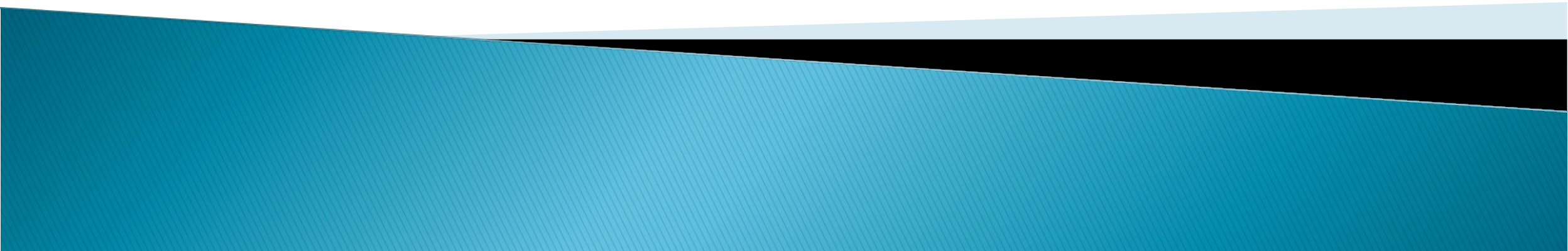
Choose **Data » Data Tools » Data Validation**.

Excel displays its Data Validation dialog box having 3 tabs-

- **Settings**
 - **Input Message**
 - **Error alert.**
- 

Settings

Here you can set the type of validation you need. Choose an option from the Allow drop-down list. The contents of the Data Validation dialog box will change, displaying controls based on your choice –

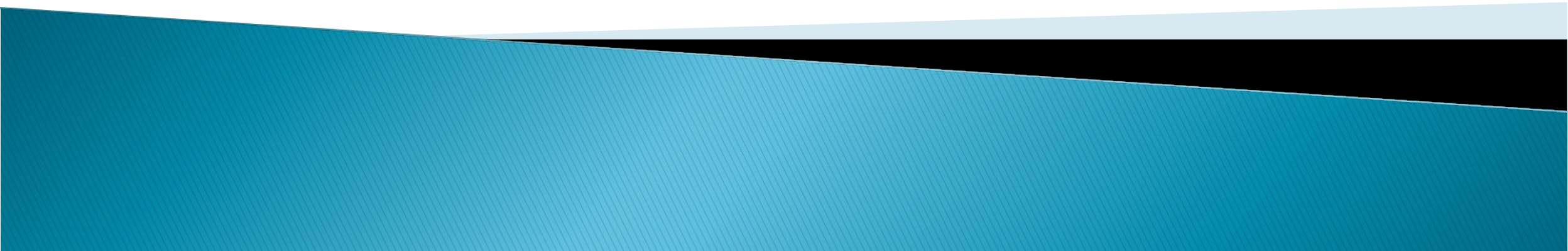
- Any Value
 - Whole Number
 - Decimal
 - List etc
- 

Circle Invalid data

We can mark invalid data by circle by using following steps:

- First take a list of data in a column.
 - Select the data, go to “Data validation” tab for setting the conditions.
 - Then click to ‘Circle invalid data’ option and invalid data will be circled.
- 

Formula



Formula

A formula, by which we can do some calculation and other operations. It returns a result, which is displayed in that cell only, where the formula is written.



Creating Formula

- ▶ For creating formula you need to type in Formula Bar. Formula begins with '=' sign.
 - ▶ When building formulas manually, you can either type the cell addresses or you can select them in the worksheet.
 - ▶ Using the selecting method to supply the cell address for formulas is often easier and more powerful method of formula building.
- 

Cell address

Every cell in Excel has an address. A cell address is a combination of Column name followed by the row number. For an example 'A1' / 'C2' etc.

There are two types of cell addresses: **relative** and **absolute**. By default, all **cell** address have **relative** address.



Relative cell address

By default, all **cell** addresses are **relative** addresses. When copied across multiple **cells**, they change based on the **relative** position of rows and columns.

if you copy the **formula** =A1+B1 from row 1 to row 2, the **formula** will become =A2+B2



Absolute Cell Address

The row and column references do not change when you copy the formula because the reference is to an actual cell address. An absolute reference uses two dollar signs in its address: one for the column letter and one for the row number (for example, \$A\$5).



Drawbacks of Formula

A user defined function called by a formula in a worksheet cell, cannot change the environment of Microsoft excel. This Means that, such a function cannot do any of the following:

- ▶ Insert, Delete, Format Cells on the spreadsheet.
 - ▶ Change another Cell's value.
 - ▶ Move, Rename, Delete or add sheets to a workbook.
 - ▶ Change any of the environment option such as calculation mode or screen views.
 - ▶ Add Names to a workbook.
 - ▶ Set properties or execute most methods.
- 

Conditional Formatting

Conditional formatting is used to format a cell or a range of cells by user defined criteria.

- ▶ Choose Home Tab » Style group » Conditional Formatting dropdown.

Conditional Formatting

- New Rule: It opens the New Formatting Rule dialog box, where you define a custom conditional formatting rule to apply to the cell selection.
- Clear Rule: It opens a continuation menu, where you can remove conditional formatting rules for the cell selection by clicking the Selected Cells option, for the entire worksheet by clicking the **‘Entire Sheet’** option, or for just the current data table by clicking the **‘This Table’** option.
- Manage Rule: It opens the Conditional Formatting Rules Manager dialog box, where you edit and delete particular rules as well as adjust their rule precedence by moving them up or down in the Rules list box.

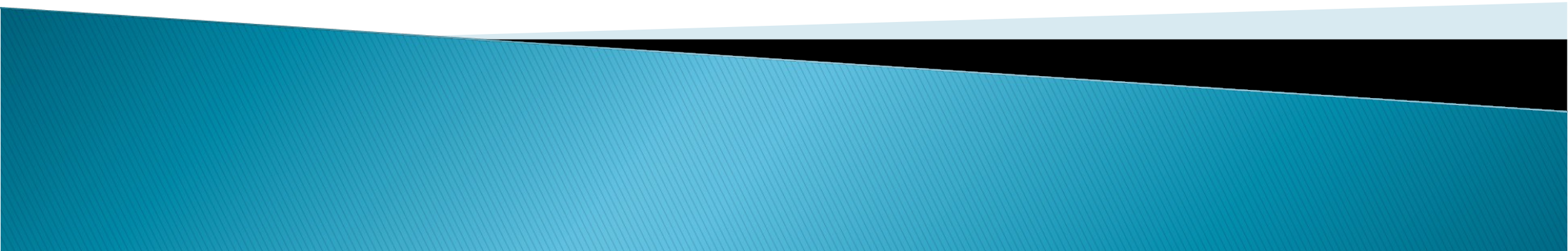
V-LOOKUP & H-LOOKUP

V-LOOKUP is a function, which is used to search for value vertically from lookup table.

Similarly H-LOOKUP searches for value horizontally from lookup table.

V-LOOKUP and H-LOOKUP both have 4 parameters as below.

(lookup_value, table_array, col_index_num, range_lookup)

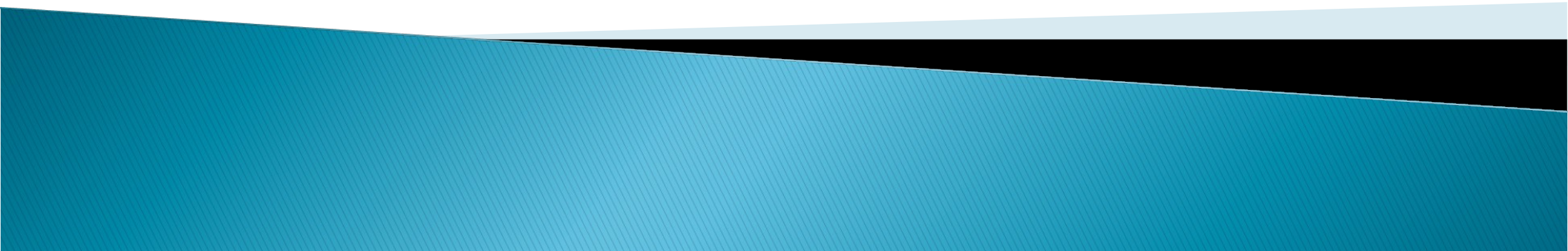


V-LOOKUP Parameters

lookup_value : It is the user input. This is the value that the function uses to search on.

The table_array : It is the area of cells in which the table is located. This includes not only the column being searched on, but the data columns for which you are going to get the values that you need.

Col_index_num : It is the column of data that contains the answer that you want.



V-LOOKUP Parameter

Range_lookup : It is a TRUE or FALSE value. When set to TRUE, the lookup function gives the closest match to the lookup_value without going over the lookup_value. When set to FALSE, an exact match must be found to the lookup_value or the function will return #N/A. Note, this requires that the column containing the lookup_value be formatted in ascending order.



Pivot Tables in Excel



Pivot Table

- In the row and column label areas of a PivotTable report, you can group the items in a field by customize ways.
 - Grouping data can help you to create a subset of data, which cannot be easily grouped in other ways, such as sorting and filtering.
- 

Pivot Table

What can we do with Pivot Table?

- We can do dimension wise grouping

What to show in a group?

- Any possible dynamic summary of the group



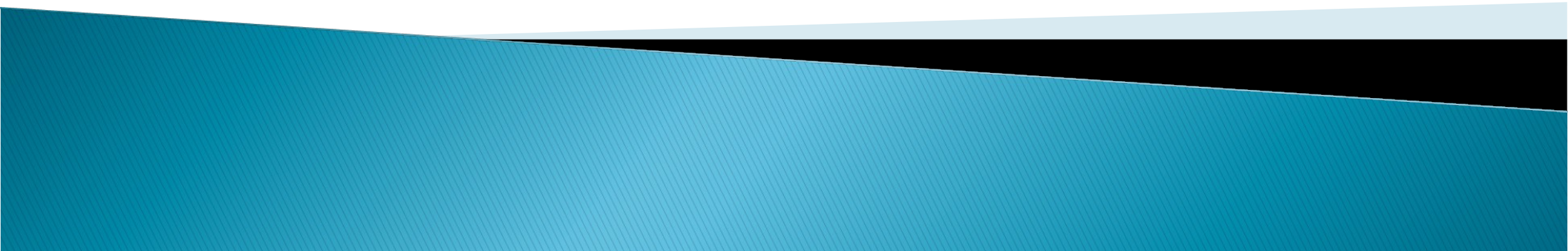
Pivot Charts Excel



Pivot Charts

A pivot chart is a graphical representation of a data summary displayed in a pivot table. A pivot chart is always based on a pivot table. Although Excel lets you create a pivot table and a pivot chart at the same time, you can't create a pivot chart without a pivot table. All Excel charting features are available in a pivot chart.

Pivot charts are available under **Insert tab » PivotTable dropdown » PivotChart**



Simple Charts in Excel



Charts

We can show the relation between two columns having numeric data by plotting charts, which have numeric data of one column along X axis and other column's numeric data of along Y axis.

Graph or Chart translate the data into distances and plot data points in a way that their relative distances are kept.

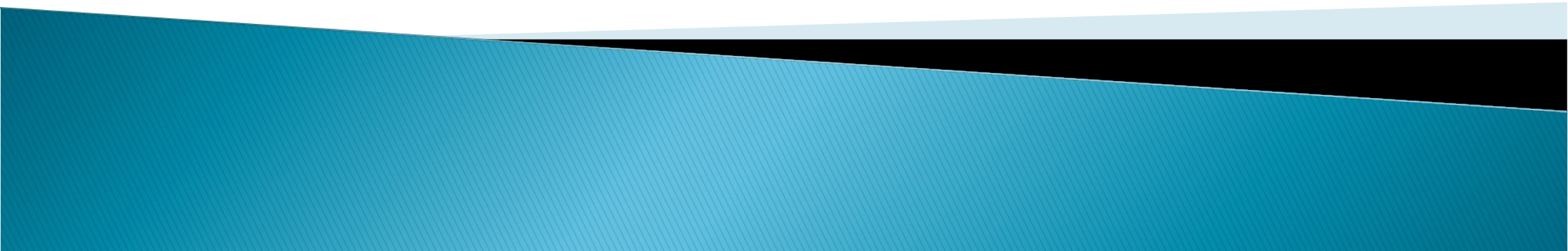


Charts using more than Two columns

If we have 3 columns of different numeric data, we can show the relation of the data by plotting data of one column along x-axis and other two columns along Y-axis(y1,y2) to create a chart that shows the visual .



Creating chart

- To create charts for the data by below steps.
 - Select the data for which you want to create chart.
 - Choose **Insert Tab » Select the chart or click on the Chart group** to see various chart types.
 - Select the chart of your choice and click OK to generate the chart.
- 

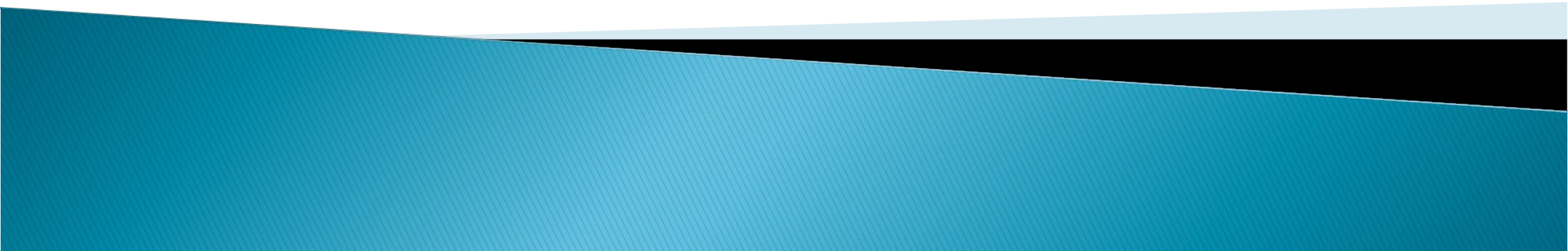
Types of Charts

Column: Column chart shows data changes over a period of time or illustrates comparisons among items.

Bar : A bar chart illustrates comparisons among individual items

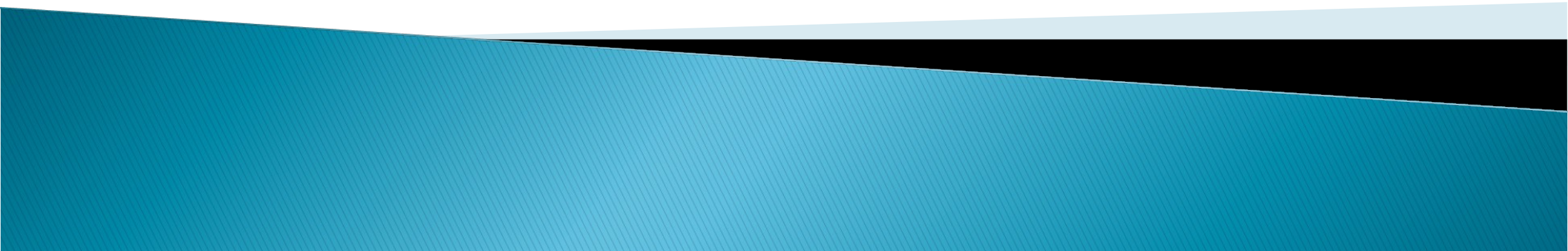
Line : A line chart shows trends in data at equal intervals.

Area : An area chart emphasizes the magnitude of change over time.



Types of Charts

Pie: A pie chart shows the size of items that make up a data series, proportional to the sum of the items. It always shows only one data series and is useful when you want to emphasize a significant element in the data.



End of the slides

Thanks!

